

Last login: Mon Jun 29 09:51:09 on ttys001
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch-200mb-2
psql (18.4)
Type "help" for help.

tpch-200mb-2=# SET ROLE alice;

EXPLAIN (ANALYZE, VERBOSE, BUFFERS)

```
select
  o_orderpriority,
  count(*) as order_count
from
  orders
where
  o_orderdate >= date '1994-01-01'
  and o_orderdate < date '1994-01-01' + interval '3' month
  and exists (
    select
      *
    from
      lineitem_secure
    where
      l_orderkey = o_orderkey
      and l_commitdate < l_receiptdate
  )
group by
  o_orderpriority
order by
  o_orderpriority;
```

SET

WARNING: Log 1: Planner Input:

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtables (
 {RANGETBLENTY
 :alias <>
 :eref
 {ALIAS
 :aliasname orders
 :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
 }
 :rtekind 0
 :relid 18182
 :inh true
 :relkind r
 :rellockmode 1
 :perminfoindex 1
 :tablesample <>
 :lateral false
 :inFromCl true
 :securityQuals <>
 }
 {RANGETBLENTY
 :alias <>
 :eref
 {ALIAS
 :aliasname *GROUP*
 :colnames ("o_orderpriority")
 }
 :rtekind 9
:groupexprs (
 {VAR
 :varno 1
 :varattno 6
 :vartype 1042
 :vartypmod 19
 :varcollid 100
 :varnullingrels (b)
 :varlevelsup 0
 :varreturningtype 0
 :varnosyn 1
 :varattnosyn 6
 :location 51
 }
)
:lateral false
:inFromCl false
:securityQuals <>
}
:rteperminfos (
 {RTEPERMISSIONINFO
 :relid 18182
 :inh true
 :requiredPerms 2
 :checkAsUser 0
 :selectedCols (b 8 12 13)
 :insertedCols (b)
 :updatedCols (b)
 }
)
}
:jointree
{FROMEXPR
:fromlist (
 {RANGETBLREF
 :rtindex 1
 }
)
}
:quals
{BOOLEXP
:boolop and
:args (
 {OPEXP
 :opno 1098
 :opfuncid 1090
 :opresulttype 16
 :opretset false
 :opcollid 0
 :inputcollid 0
 :args (
 {VAR
 :varno 1
 :varattno 5
 :vartype 1082
 :vartypmod -1
 :varcollid 0
 :varnullingrels (b)
 :varlevelsup 0
 :varreturningtype 0
 :varnosyn 1
 :varattnosyn 5
 :location 134
 }
 {CONST
 :consttype 1082
 :consttypmod -1
 :constcollid 0
 :constlen 4
 :constbyval true
 :constisnull false
 :location 154

o_orderdate >= DATE '1994-01-01'

```

:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
:location 146
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opresultset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varstype 1082
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 179
}
{OPEXPR
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opresultset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 198
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:conststypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 222
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
)
:location 211
}
)
:location 191
}
{SUBLINK
:sublinktype 0
:sublinkid 0
:restexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGEBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem_secure
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGEBLENTTRY
:alias
{ALIAS
:aliasname l
:colnames <>
}
}
:eref
{ALIAS
:aliasname l
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_
linenumber" "l_quantity" "l_extendedprice" "l_disco
unt" "l_tax" "l_returnflag" "l_linestatus" "l_shipd
ate" "l_commitdate" "l_receiptdate" "l_shipinstruct
" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 18194
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rtperminfos (
{RTEPERMISSIONINFO
:relid 18194
:inh true

```

o_orderdate < DATE '1994-01-01' + INTERVAL '3' MONTH

```

:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20
21 22 23)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{SUBLINK
:sublinkType 0
:subLinkId 0
:testexpr <>
:opName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rttable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_
orderstatus" "o_totalprice" "o_orderdat
e" "o_orderpriority" "o_clerk" "o_shipp
riority" "o_comment")
}
:rtkind 0
:relid 18182
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rtperminfos (
{RTEPERMISSIONINFO
:relid 18182
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
}
:location -1
}
)
{SCALARARRAYOPEXP
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 6
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6

```

o.o_orderkey =
outer.o_orderkey

```

        :location -1
      }
      {ARRAYEXPR
      :array_typeid 1014
      :array_collid 100
      :element_typeid 1042
      :elements (
      {CONST
      :consttype 1042
      :consttypmod -1
      :constcollid 100
      :constlen -1
      :constbyval false
      :constisnull false
      :location -1
      :constvalue 12 [ 48 0 0 0 49 4
      5 85 82 71 69 78 84 ]
      }
      {CONST
      :consttype 1042
      :consttypmod -1
      :constcollid 100
      :constlen -1
      :constbyval false
      :constisnull false
      :location -1
      :constvalue 10 [ 40 0 0 0 50 4
      5 72 73 71 72 ]
      }
      )
      :multidims false
      :list_start -1
      :list_end -1
      :location -1
    }
  }
  :location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
)
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "p
s_availqty" "ps_supplycost" "ps_comment
")
}
:rtkind 0
:relid 18163
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18163
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10)
:insertedCols (b)

```

o_orderpriority IN ('1-URGENT', '2-HIGH')

```

      :updatedCols (b)
    }
  }
}
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
  :rtindex 1
  }
)
:quals
{BOOLEXP
:boolop and
:args (
  {OPEXP
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
    }
    {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
    }
  )
  :location -1
  }
  {OPEXP
  :opno 96
  :opfuncid 65
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
    }
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
  )
  :location -1
  }
  {OPEXP
  :opno 521
  :opfuncid 147
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
    }
    {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 0 0 0 0 0 0 0 ]
    }
  )
  :location -1
  }
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
  :expr
  {CONST
  :consttype 23
  :consttypmod -1
  :constcollid 0
  :constlen 4
  :constbyval true
  :constisnull false
  :location -1
  :constvalue 4 [ 1 0 0 0 0 0 0 ]
  }
  :resno 1
  :resname ?column?
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false

```

```

ps_partkey = l_partkey
AND ps_suppkey =
l_suppkey
AND ps_availqty > 0

```

```

    }
  }
  :override 0
  :onConflict <>
  :returningOldAlias <>
  :returningNewAlias <>
  :returningList <>
  :groupClause <>
  :groupDistinct false
  :groupingSets <>
  :havingQual <>
  :windowClause <>
  :distinctClause <>
  :sortClause <>
  :limitOffset <>
  :limitCount <>
  :limitOption 0
  :rowMarks <>
  :setOperations <>
  :constraintDeps <>
  :withCheckOptions <>
  :stmt_location -1
  :stmt_len -1
  }
  :location -1
  }
  :location -1
  }
  }
  :mergeActionList <>
  :mergeTargetRelation 0
  :mergeJoinCondition <>
  :targetList (
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 1
  :vartype 23
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 1
  :location -1
  }
  :resno 1
  :resname l_orderkey
  :ressortgroupref 0
  :resorigtbl 18194
  :resorigcol 1
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 2
  :vartype 23
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 2
  :location -1
  }
  :resno 2
  :resname l_partkey
  :ressortgroupref 0
  :resorigtbl 18194
  :resorigcol 2
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 3
  :vartype 23
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 3
  :location -1
  }
  :resno 3
  :resname l_suppkey
  :ressortgroupref 0
  :resorigtbl 18194
  :resorigcol 3
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 4
  :vartype 23
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 4
  :location -1
  }
  :resno 4
  :resname l_linenum
  :ressortgroupref 0
  :resorigtbl 18194
  :resorigcol 4
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 5
  :vartype 1700
  :vartypmod 983846
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 5
  :location -1
  }
  :resno 5
  :resname l_quantity
  :ressortgroupref 0
  :resorigtbl 18194
  :resorigcol 5
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 1

```

```

:varattno 6
:vartype 1700
:vartypmod 983846
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname 1_extendedprice
:resortgroupref 0
:resorigtbl 18194
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983846
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
}
:resno 7
:resname 1_discount
:resortgroupref 0
:resorigtbl 18194
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983846
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
}
:resno 8
:resname 1_tax
:resortgroupref 0
:resorigtbl 18194
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
}
:resno 9
:resname 1_returnflag
:resortgroupref 0
:resorigtbl 18194
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
}
:resno 10
:resname 1_linestatus
:resortgroupref 0
:resorigtbl 18194
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
}
:resno 11
:resname 1_shipdate
:resortgroupref 0
:resorigtbl 18194
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
}
:resno 12
:resname 1_commitdate
:resortgroupref 0
:resorigtbl 18194
:resorigcol 12

```

```

:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 1042
:vartypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location -1
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 1043
:vartypmod 48
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location -1
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier true
:relid 18252
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rtoperinfos (
{RTEPERMISSIONINFO
:relid 18252
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP

```

```

:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 409
  }
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 422
  }
)
:location 420
}
{OPEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 12
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 12
    :location 461
  }
  {VAR
    :varno 1
    :varattno 13
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 13
    :location 476
  }
)
:location 474
}
)
:location 457
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location 300
    }
  }
  :resno 1
  :resname l_orderkey
  :ressortgroupref 0
  :resorigtbl 18252
  :resorigcol 1
  :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location 300
    }
  }
  :resno 2
  :resname l_partkey
  :ressortgroupref 0
  :resorigtbl 18252
  :resorigcol 2
  :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 3
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location 300
    }
  }
  :resno 3
  :resname l_supkey
  :ressortgroupref 0
  :resorigtbl 18252
  :resorigcol 3
  :resjunk false
  }
  {TARGETENTRY

```

l_orderkey =
o_orderkey
AND l_commitdate <
l_receiptdate

```

:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 300
}
:resno 4
:resname l_linenummer
:ressortgroupref 0
:resorigtbl 18252
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 300
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 18252
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 300
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 18252
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 300
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 18252
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 300
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 18252
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 300
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 18252
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 300
}
:resno 10
:resname l_linestatus

```

```

:ressortgroupref 0
:resorigtbl 18252
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 300
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 18252
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 300
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 18252
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 300
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 18252
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 300
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 18252
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 1042
:vartypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 300
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 18252
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 1043
:vartypmod 48
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location 300
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 18252
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>

```

```

        :distinctClause <>
        :sortClause <>
        :limitOffset <>
        :limitCount <>
        :limitOption 0
        :rowMarks <>
        :setOperations <>
        :constraintDeps <>
        :withCheckOptions <>
        :stmt_location 0
        :stmt_len 0
    }
    :location 244
}
)
:location 175
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 1042
:vartypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 18182
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggnosyn 1
:aggtransno -1
:location 76
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 1054
:sortop 1058
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 18182
:inh false
}

```

```

:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:varatype 1042
:varattnum 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 51
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem_secure
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname l
:colnames <>
}
}
:eref
{ALIAS
:aliasname l
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}
:rtekind 0
:relid 18194
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18194
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname o
:colnames <>
}
}
:eref
{ALIAS
:aliasname o

```

```

EXISTS (
  SELECT 1
  FROM orders o
  WHERE
    o.o_orderkey =
    l.l_orderkey
    AND
    o.o_orderpriority IN ('1-
    URGENT', '2-HIGH')
)

```

```

:colnames ("o_orderkey" "o_custkey" "o_orderstatus"
"o_totalprice" "o_orderdate" "o_orderpriority" "o_
clerk" "o_shippriority" "o_comment")
}
}
:rtekind 0
:relid 18182
:inh true
:relkind r
:rellockmode 1
:petminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18182
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromList (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
{VAR
:varno 1
:varattno 1
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
:location -1
}
)
{SCALARARRAYOEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:neqfuncid 0
:useOr true
:inputcollid 100
:args (
{VAR
:varno 1
:varattno 6
:varstype 1042
:varstypmod 19
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
)
{ARRAYEXPR
:array_typeid 1014
:array_collid 100
:element_typeid 1042
:elements (
{CONST
:consttype 1042
:conststypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 12 [ 48 0 0 0 49 45 85 82 71 6
9 78 84 ]
}
{CONST
:consttype 1042
:conststypmod -1
:constcollid 100
:constlen -1
:constbyval false
:constisnull false
:location -1
:constvalue 10 [ 40 0 0 0 50 45 72 73 71 7
2 ]
}
)
:multidims false
:list_start -1
:list_end -1
:location -1
}
)
:location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
}
}
)

```

```

:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rttable (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "ps_availability"
"ps_supplycost" "ps_comment")
}
:rtekind 0
:relid 18163
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18163
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
}
:location -1
}
)
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16

```

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 2
    :location -1
  }
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 1
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
)
:location -1
}
}
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 3
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 3
    :location -1
  }
  {CONST
    :consttype 23
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
  }
)
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname l_orderkey
    :ressortgroupref 0
    :resorigtbl 18194
    :resorigcol 1
    :resjunk false
  }
)

```

```

EXISTS (
  SELECT 1
  FROM partsupp ps
  WHERE
ps.ps_partkey =
l.l_partkey
  AND
ps.ps_suppkey =
l.l_suppkey
  AND ps.ps_availqty
> 0
)

```

```

}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname l_supkey
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname l_linenummer
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
}

```

```

:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 1042
:varypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:varypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:varypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:varypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:varypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:varypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 1042
:varypmod 14
:varcollid 100
:varnullingrels (b)

```

```

        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 15
        :location -1
    }
    :resno 15
    :resname l_shipmode
    :ressortgroupref 0
    :resorigtbl 18194
    :resorigcol 15
    :resjunk false
    }
    {TARGETENTRY
    :expr
        {VAR
        :varno 1
        :varattno 16
        :vartype 1043
        :vartypmod 48
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 16
        :location -1
        }
    :resno 16
    :resname l_comment
    :ressortgroupref 0
    :resorigtbl 18194
    :resorigcol 16
    :resjunk false
    }
    }
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location -1
    :stmt_len -1
    }
    :security_barrier true
    :relid 18252
    :inh false
    :relkind v
    :rellockmode 1
    :perminfoindex 2
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    }
    :rteperminfos (
    {RTEPERMISSIONINFO
    :relid 18182
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 12 13)
    :insertedCols (b)
    :updatedCols (b)
    }
    {RTEPERMISSIONINFO
    :relid 18252
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
    :insertedCols (b)
    :updatedCols (b)
    }
    )
    :jointree
    {FROMEXPR
    :fromlist (
    {JOINEXPR
    :jointype 4
    :isNatural false
    :larg
    {FROMEXPR
    :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
    )
    :quals (
    {OEXPR
    :opno 1098
    :opfuncid 1090
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 134
    }
    {CONST
    :consttype 1082
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 154
    :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
    }
    )
    :location 146
    }
    {OEXPR
    :opno 2345
    :opfuncid 2338
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 1
    :varattno 5

```

FROM orders o

SEMI JOIN lineitem l
ON l.l_orderkey =
o.o_orderkey
AND l.l_commitdate <
l.l_receiptdate

WHERE
o.o_orderdate >=
DATE '1994-01-01'
AND o.o_orderdate <
DATE '1994-01-01' +
INTERVAL '3 month'

```

        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 5
        :location 179
    }
    {CONST
    :consttype 1114
    :consttypmod -1
    :constcollid 0
    :constlen 8
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
    }
    }
    :location 191
    }
    }
    :rarg
    {RANGETBLREF
    :rtindex 3
    }
    :usingClause <>
    :join_using_alias <>
    :quals (
    {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 3
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 1
    :location 409
    }
    {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 422
    }
    )
    :location 420
    }
    {OPEXPR
    :opno 1095
    :opfuncid 1087
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 3
    :varattno 12
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 12
    :location 461
    }
    {VAR
    :varno 3
    :varattno 13
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 13
    :location 476
    }
    )
    :location 474
    }
    )
    :alias <>
    :rtindex 0
    }
    )
    :quals <>
    )
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 6
    :vartype 1042
    :vartypmod 19
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 10182
    :resorigcol 6
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {AGGREF
    :aggfnoid 2803
    :aggtype 20
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes <>

```

```

:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggregatesorted false
:aggrelevsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 76
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 1054
    :sortop 1058
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 1054
    :sortop 1058
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:serialOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1098 :opfuncid 1090 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 5 :vartype 10 82 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1) {CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [113 -9 -1 -1 -1 -1 -1 -1]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :router_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1098 :opfuncid 1090 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 1 :varattno 5 :vartype 1 802 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1) {CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [113 -9 -1 -1 -1 -1 -1 -1]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :router_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varattno 6 :vartype 1082 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) (VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)

WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varattno 6 :vartype 1082 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1) (VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) | parallel_safe = true | reason = PROPAGATE_SAFE

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 1095 :opfuncid 1087 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({(VAR :varno 3 :varattno 12 :vartype 1 802 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 12 :location -1) {VAR :varno 3 :varattno 13 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 13 :location -1}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b 3) :left_relds (b 3) :right_relds (b 3) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :router_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 3 :varattno 12 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 12 :location -1) {VAR :varno 3 :varattno 13 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 13 :location -1} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b 3) :left_relds (b 3) :right_relds (b 3) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :router_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mvcfreq -1 :right_mvcfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPAGATE_SAFE

WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1} | parallel_safe = true | reason = PROPAGATE_SAFE

Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:table (
 {RANGETBLENTY
:alias
{ALIAS
:aliasname 1
:colnames <>
}
:eref
{ALIAS
:aliasname 1
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:reld 10194
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>

```

:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtekind 0
:relid 18182
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY
:alias
{ALIAS
:aliasname ps
:colnames <>
}
}
:eref
{ALIAS
:aliasname ps
:colnames ("ps_partkey" "ps_suppkey" "ps_availqty" "ps_supplycost" "ps_comment")
}
}
:rtekind 0
:relid 18163
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
}
rteperminfos (
{RTEPERMISSIONINFO
:relid 18194
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18182
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 18163
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10)
:insertedCols (b)
:updatedCols (b)
}
}
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals (
{OPEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
)
{VAR
:varno 1
:varattno 13
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
)
:location 474
}
)
}
:rarg
{RANGETBLREF
:rtindex 2
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16

```

l_commitdate <
l_receiptdate

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
  }
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location -1
  }
)
:location -1
}
{SCALARARRAYOPEXPR
:opno 1054
:opfuncid 1048
:hashfuncid 0
:negfuncid 0
:useOr true
:inputcollid 100
:args (
  {VAR
    :varno 2
    :varattno 6
    :vartype 1042
    :vartypmod 19
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 6
    :location -1
  }
  {CONST
    :consttype 1014
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location -1
    :constvalue 40 [ -64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0
1 0 0 0 40 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 7
3 71 72 0 0 ]
  }
)
:location -1
}
)
:alias <>
:rtindex 0
}
:rarg
{RANGETBLREF
:rtindex 3
}
:usingClause <>
:join_using_alias <>
:quals (
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 3
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 1
        :location -1
      }
      {VAR
        :varno 1
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 2
        :location -1
      }
    )
  }
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 3
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 2
        :location -1
      }
      {VAR
        :varno 1
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1

```

o_orderkey =
l_orderkey
AND o_orderpriority IN
('1-URGENT', '2-HIGH')

ps_partkey = l_partkey
AND ps_suppkey =
l_suppkey
AND ps_availqty > 0

```

        :varattnosyn 3
        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 3
        :varattno 3
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 3
        :location -1
    }
    {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 4 [ 0 0 0 0 0 0 0 0 ]
    }
)
:location -1
}
)
:alias <>
:rtindex 0
}
)
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {VAR
            :varno 1
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 1
            :varattnosyn 1
            :location -1
        }
        :resno 1
        :resname l_orderkey
        :ressortgroupref 0
        :resorigtbl 18194
        :resorigcol 1
        :resjunk false
    }
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull true
            :location -1
            :constvalue <>
        }
        :resno 2
        :resname l_partkey
        :ressortgroupref 0
        :resorigtbl 18194
        :resorigcol 2
        :resjunk false
    }
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull true
            :location -1
            :constvalue <>
        }
        :resno 3
        :resname l_supkey
        :ressortgroupref 0
        :resorigtbl 18194
        :resorigcol 3
        :resjunk false
    }
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull true
            :location -1
            :constvalue <>
        }
        :resno 4
        :resname l_linenum
        :ressortgroupref 0
        :resorigtbl 18194
        :resorigcol 4
        :resjunk false
    }
    {TARGETENTRY
        :expr
        {CONST
            :consttype 1700
            :consttypmod 983046
            :constcollid 0
            :constlen -1
            :constbyval false
            :constisnull true
            :location -1
            :constvalue <>
        }
        :resno 5
        :resname l_quantity
        :ressortgroupref 0
        :resorigtbl 18194
        :resorigcol 5
        :resjunk false
    }
)

```

```

}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 6
:resname l_extendedprice
:resortgroupref 0
:resorigtbl 18194
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 7
:resname l_discount
:resortgroupref 0
:resorigtbl 18194
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1700
:consttypmod 983046
:constcollid 0
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 8
:resname l_tax
:resortgroupref 0
:resorigtbl 18194
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 5
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 9
:resname l_returnflag
:resortgroupref 0
:resorigtbl 18194
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 5
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 10
:resname l_linestatus
:resortgroupref 0
:resorigtbl 18194
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 11
:resname l_shipdate
:resortgroupref 0
:resorigtbl 18194
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 12
:resname l_commitdate
:resortgroupref 0
:resorigtbl 18194
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull true
:location -1
:constvalue <>
}
:resno 13
:resname l_receiptdate

```

```
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 29
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1042
:consttypmod 14
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{CONST
:consttype 1043
:consttypmod 48
:constcollid 100
:constlen -1
:constbyval false
:constisnull true
:location -1
:constvalue <>
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 18194
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICINFO :clause {OEXPR :opno 1095 :opfuncid 1087 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varrattno 12 :vartype 1 082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 13 :location -1}} {VAR :varno 1 :varrattno 13 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 13 :location -1}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b 1) :left_rels (b 1) :right_rels (b 1) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {OEXPR :opno 1095 :opfuncid 1087 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varrattno 12 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 13 :location -1}} {VAR :varno 1 :varrattno 13 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 13 :location -1}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 1) :required_rels (b 1) :incompatible_rels (b) :outer_rels (b 1) :left_rels (b 1) :right_rels (b 1) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1} {VAR :varno 1 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 2 :location -1} {VAR :varno 1 :varrattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1} {VAR :varno 1 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 2 :location -1} {VAR :varno 1 :varrattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 3 :location -1})
WARNING: is_parallel_safe: INPUT qual = ({RESTRICINFO :clause {SCALARARRAYOEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args {(VAR :varno 2 :varrattno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 6 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 49 45 85 82 71 69 78 84 40 0 0 50 45 72 73 71 72 0 0]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {SCALARARRAYOEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args {(VAR :varno 2 :varrattno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 6 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 48 [-64 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 49 45 85 82 71 69 78 84 40 0 0 50 45 72 73 71 72 0 0]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 2) :required_rels (b 2) :incompatible_rels (b) :outer_rels (b) :left_rels (b) :right_rels (b) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 1 :location -1} {VAR :varno 2 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 2 :location -1} {VAR :varno 2 :varrattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 3 :location -1})
WARNING: is_parallel_safe: INPUT qual = ({RESTRICINFO :clause {OEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 3 :varrattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [0 0 0 0 0 0 0 0]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 3) :required_rels (b 3) :incompatible_rels (b) :outer_rels (b 3) :left_rels (b 3) :right_rels (b 3) :orclause <> :rinfo_serial 6 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {OEXPR :opno 521 :opfuncid 147 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 3 :varrattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 3 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [0 0 0 0 0 0 0 0]}} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_rels (b 3) :required_rels (b 3) :incompatible_rels (b) :outer_rels (b 3) :left_rels (b 3) :right_rels (b 3) :orclause <> :rinfo_serial 6 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 0 :right_hashoperator 0})
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 3 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 1 :location -1} {VAR :varno 3 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 2 :location -1} {VAR :varno 3 :varrattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 3 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 3 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 1 :location -1} {VAR :varno 3 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 2 :location -1} {VAR :varno 3 :varrattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varrattnosyn 3 :location -1})
WARNING: is_parallel_safe: INPUT qual = ({RESTRICINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1} {VAR :varno 1 :varrattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 2 :location -1} {VAR :varno 1 :varrattno 3 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 3 :location -1}} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_rels (b 1 2) :required_rels (b 1 2) :incompatible_rels (b) :outer_rels (b 1) :left_rels (b 1) :right_rels (b 2) :orclause <> :rinfo_serial 7 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec 0.40213333333333334 :mergeofamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 1 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varrattnosyn 1 :location -1} :em_rels (b 1) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2 3)}} :right_em (EQUIVALENCEMEMBER :em_expr {VAR :varno 2 :varrattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varrattnosyn 1 :location -1} :em_rels (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_rels (b 1 2 3)}} :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96})
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args {(VAR :varno 1 :varrattno 1 :vartype 23 :v
```

[illegible]

```

line 29: constcollid 100 :constlen -1 :constbyval true :constnull true :location -1 :constvalue <> <> (CONST :consttype 1842 :consttypemod 14 :constcollid 100 :constlen -1 :constbyval false :constnull
true :location -1 :constvalue <>) (CONST :consttype 1843 :consttypemod 48 :constcollid 100 :constlen -1 :constbyval false :constnull true :location -1 :constvalue <>))
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1)
) (CONST :consttype 23 :consttypemod 10 :constcollid 0 :constlen 4 :constbyval true :constnull true :location -1 :constvalue <>) (CONST :consttype 23 :consttypemod 11 :constcollid 0 :constlen 4 :constby
val true :constnull true :location -1 :constvalue <>) (CONST :consttype 23 :consttypemod -1 :constcollid 0 :constlen 4 :constbyval true :constnull true :location -1 :constvalue <>) (CONST :consttype
23 :consttypemod -1 :constcollid 0 :constlen 4 :constbyval true :constnull true :location -1 :constvalue <>) (CONST :consttype 1700 :consttypemod 983046 :constcollid 0 :constlen -1 :constbyval false :constnull true :location -1 :constvalue <>) (CONST :constt
ype 1700 :consttypemod 983046 :constcollid 0 :constlen -1 :constbyval false :constnull true :location -1 :constvalue <>) (CONST :consttype 1842 :consttypemod 5 :constcollid 100 :constlen -1 :constbyval
false :constnull true :location -1 :constvalue <>) (CONST :consttype 1842 :consttypemod 5 :constcollid 100 :constlen -1 :constbyval false :constnull true :location -1 :constvalue <>) (CONST :constt
ype 1882 :consttypemod -1 :constcollid 0 :constlen 4 :constbyval true :constnull true :location -1 :constvalue <>) (CONST :consttype 1882 :consttypemod -1 :constcollid 0 :constlen 4 :constbyval true :cons
tnull true :location -1 :constvalue <>) (CONST :consttype 1882 :consttypemod -1 :constcollid 0 :constlen 4 :constbyval true :constnull true :location -1 :constvalue <>) (CONST :consttype 1842 :constt
ypemod 29 :constcollid 100 :constlen -1 :constbyval false :constnull true :location -1 :constvalue <>) (CONST :consttype 1842 :consttypemod 14 :constcollid 100 :constlen -1 :constbyval false :constnul
l true :location -1 :constvalue <>) (CONST :consttype 1843 :consttypemod 48 :constcollid 100 :constlen -1 :constbyval false :constnull true :location -1 :constvalue <>)) | parallel_safe = true | reason
= PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = (VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location
-1) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = (VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location
-1) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((RESTRICTINFO :clause (OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 1 :vartype 23 :va
rtypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 3 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 3) :required_relds (b 1 3) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 3) :orclause <> :rinfo_s
erial 5 :eval_cost_startup 1 :eval_cost_per_tuple 2 :norm_select 0.5 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3)) :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right
_bucketsize -1 :left_mcvfreq 1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96))
WARNING: is_parallel_safe: OUTPUT qual = ((RESTRICTINFO :clause (OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ((VAR :varno 1 :varattno 1 :vartype 23 :v
artypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) (VAR :varno 3 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (
b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1) :location -1) :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false
:has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 1 3) :required_relds (b 1 3) :incompatible_relds (b) :outer_relds (b 1) :right_relds (b 3) :orclause <> :rinfo_s
erial 5 :eval_cost_startup 1 :eval_cost_per_tuple 2 :norm_select 0.5 :mergeopfamilies (o 1976) :left_em (EQUIVALENCEMEMBER :em_expr (VAR :varno 1 :varattno 1 :vartype 23 :vartypemod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1) :em_relds (b 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain (JOINDOMAIN :jd_relds (b 1 3)) :outer_is_left false :hashjoinoperator 96 :left_bucketsize -1 :right
_bucketsize -1 :left_mcvfreq 1 :right_mcvfreq -1 :left_hashoperator 96 :right_hashoperator 96)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location
-1) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :locatio
n -1) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location
-1) {AGGREG :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic fals
e :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :locatio
n -1) {AGGREG :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic fals
e :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location
-1) {AGGREG :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic fals
e :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :locatio
n -1) {AGGREG :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic fals
e :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location
-1) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location
-1) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :locatio
n -1) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location
-1) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location
-1) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: OUTPUT qual = ((VAR :varno 1 :varattno 6 :vartype 1842 :vartypemod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :locatio
n -1) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ((VAR :varno
```

```

GroupAggregate (cost=72382.83..72425.95 rows=25 width=24) (actual time=332.810..333.586 rows=2.00 loops=1)
  Output: orders.o_orderpriority, count(*)
  Group Key: orders.o_orderpriority
  Buffers: shared hit=1880 read=34763 written=1
  -> Sort (cost=72382.83..72397.19 rows=5742 width=16) (actual time=332.798..333.296 rows=4183.00 loops=1)
    Output: orders.o_orderpriority
    Sort Key: orders.o_orderpriority
    Sort Method: quicksort Memory: 193kB
    Buffers: shared hit=1880 read=34763 written=1
    -> Hash Right Semi Join (cost=23279.13..72824.32 rows=5742 width=16) (actual time=130.473..332.217 rows=4183.00 loops=1)
      Output: orders.o_orderpriority
      Hash Cond: (lineitem_secure.l_orderkey = orders.o_orderkey)
      Buffers: shared hit=1877 read=34763 written=1
      -> Gather (cost=13840.07..61920.49 rows=168849 width=4) (actual time=73.293..246.748 rows=384150.00 loops=1)
        Output: lineitem_secure.l_orderkey
        Workers Planned: 2
        Workers Launched: 2
        Buffers: shared hit=1147 read=38221 written=1
        -> Subquery Scan on lineitem_secure (cost=12840.07..44835.59 rows=67020 width=4) (actual time=71.080..248.926 rows=101383.33 loops=3)
          Output: lineitem_secure.l_orderkey
          Buffers: shared hit=1147 read=38221 written=1
          Worker 0: actual time=69.924..259.543 rows=108004.00 loops=1
            Buffers: shared hit=292 read=10635 written=1
          Worker 1: actual time=70.260..259.527 rows=108300.00 loops=1
            Buffers: shared hit=326 read=10611
          -> Parallel Hash Semi Join (cost=12840.07..44165.39 rows=67020 width=370) (actual time=71.079..241.639 rows=101383.33 loops=3)
            Output: l1_orderkey, NULL::date, NULL::integer, NULL::integer, NULL::numeric(15,2), NULL::numeric(15,2), NULL::numeric(15,2), NULL::numeric(15,2), NULL::character(1), NULL::date, NULL::date, NULL::character(25), NULL::character(10), NULL::character varying(44)
            Hash Cond: ((l1_partkey = ps_ps_partkey) AND (l1_supkey = ps.ps_supkey))
            Buffers: shared hit=1147 read=38221 written=1
            Worker 0: actual time=69.924..251.856 rows=108004.00 loops=1
              Buffers: shared hit=292 read=10635 written=1
            Worker 1: actual time=70.260..251.821 rows=108300.00 loops=1
              Buffers: shared hit=326 read=10611
            -> Parallel Hash Semi Join (cost=7462.84..37447.76 rows=67020 width=12) (actual time=27.719..151.935 rows=101383.33 loops=3)
              Output: l1_orderkey, l1_partkey, l1_supkey
              Hash Cond: (l1_orderkey = o.o_orderkey)
              Buffers: shared hit=666 read=27158 written=1
              Worker 0: actual time=27.660..160.469 rows=108004.00 loops=1
                Buffers: shared hit=193 read=9564 written=1
              Worker 1: actual time=28.018..160.266 rows=108300.00 loops=1
                Buffers: shared hit=226 read=9588
              -> Parallel Seq Scan on public.lineitem l1 (cost=0.00..28801.84 rows=166662 width=12) (actual time=0.564..81.093 rows=252963.00 loops=3)
                Output: l1_orderkey, l1_partkey, l1_supkey, l1_linenum, l1_quantity, l1_extendedprice, l1_discount, l1_tax, l1_returnflag, l1_linestatus, l1_shipdate, l1_commitdate, l1_receiptdate, l1_shipinstruct, l1_shipmode, l1_comment
                Filter: (l1_commitdate < l1_receiptdate)
                Rows Removed by Filter: 147027
                Buffers: shared read=22552 written=1
                Worker 0: actual time=0.511..86.819 rows=269092.00 loops=1
                  Buffers: shared read=7991 written=1
                Worker 1: actual time=0.859..86.561 rows=270472.00 loops=1
                  Buffers: shared read=8036
              -> Parallel Hash (cost=6834.50..6834.50 rows=50267 width=4) (actual time=27.056..27.057 rows=40048.00 loops=3)
                Output: o.o_orderkey
                Buckets: 131072 Batches: 1 Memory Usage: 5760kB
                Buffers: shared hit=666 read=4606
                Worker 0: actual time=27.132..27.132 rows=40065.00 loops=1
                  Buffers: shared hit=193 read=1573
                Worker 1: actual time=26.908..26.908 rows=40578.00 loops=1
                  Buffers: shared hit=226 read=1552
                -> Parallel Seq Scan on public.orders o (cost=0.00..6834.50 rows=50267 width=4) (actual time=0.094..21.375 rows=40048.00 loops=3)
                  Output: o.o_orderkey
                  Filter: (o.o_orderpriority = ANY ('{1-URGENT,2-HIGH}'::bpchar[]))
                  Rows Removed by Filter: 59952
                  Buffers: shared hit=666 read=4606
                  Worker 0: actual time=0.122..21.350 rows=40065.00 loops=1
                    Buffers: shared hit=193 read=1573
                  Worker 1: actual time=0.099..21.476 rows=40578.00 loops=1
                    Buffers: shared hit=226 read=1552
                -> Parallel Hash (cost=4377.33..4377.33 rows=66608 width=8) (actual time=43.129..43.129 rows=53333.33 loops=3)

```

```
Output: ps.ps_partkey, ps.ps_suppkey
Buckets: 262144 Batches: 1 Memory Usage: 8352kB
Buffers: shared hit=481 read=3863
Worker 0: actual time=42.244..42.244 rows=52050.00 loops=1
  Buffers: shared hit=99 read=1071
Worker 1: actual time=42.231..42.231 rows=50882.00 loops=1
  Buffers: shared hit=100 read=1023
-> Parallel Seq Scan on public.partsupp ps (cost=0.00..4377.33 rows=6660 width=8) (actual time=0.010..33.470 rows=53333.33 loops=3)
  Output: ps.ps_partkey, ps.ps_suppkey
  Filter: (ps.ps_availqty > 0)
  Buffers: shared hit=481 read=3863
  Worker 0: actual time=0.012..33.154 rows=52050.00 loops=1
    Buffers: shared hit=99 read=1071
  Worker 1: actual time=0.006..32.483 rows=50882.00 loops=1
    Buffers: shared hit=100 read=1023
-> Hash (cost=9295.50..9295.50 rows=11485 width=20) (actual time=56.498..56.509 rows=11352.00 loops=1)
  Output: orders.o_orderpriority, orders.o_orderkey
  Buckets: 16384 Batches: 1 Memory Usage: 705kB
  Buffers: shared hit=730 read=4542
-> Gather (cost=1000.00..9295.50 rows=11485 width=20) (actual time=0.345..54.896 rows=11352.00 loops=1)
  Output: orders.o_orderpriority, orders.o_orderkey
  Workers Planned: 2
  Workers Launched: 2
  Buffers: shared hit=730 read=4542
  -> Parallel Seq Scan on public.orders (cost=0.00..7147.00 rows=4785 width=20) (actual time=1.155..51.919 rows=3784.00 loops=3)
    Output: orders.o_orderpriority, orders.o_orderkey
    Filter: ((orders.o_orderdate >= '1994-01-01'::date) AND (orders.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zone))
    Rows Removed by Filter: 96216
    Buffers: shared hit=730 read=4542
    Worker 0: actual time=1.899..50.979 rows=3153.00 loops=1
      Buffers: shared hit=231 read=1213
    Worker 1: actual time=1.351..50.934 rows=3271.00 loops=1
      Buffers: shared hit=280 read=1286

Planning:
  Buffers: shared hit=213 read=5
Planning Time: 11.561 ms
Execution Time: 333.927 ms
(105 rows)

tpch-200mb-2=>
```